



Bayesian Filtering Approaches for 3D Localization of Quadruped Robots on the Grand Tour Dataset

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Introduction

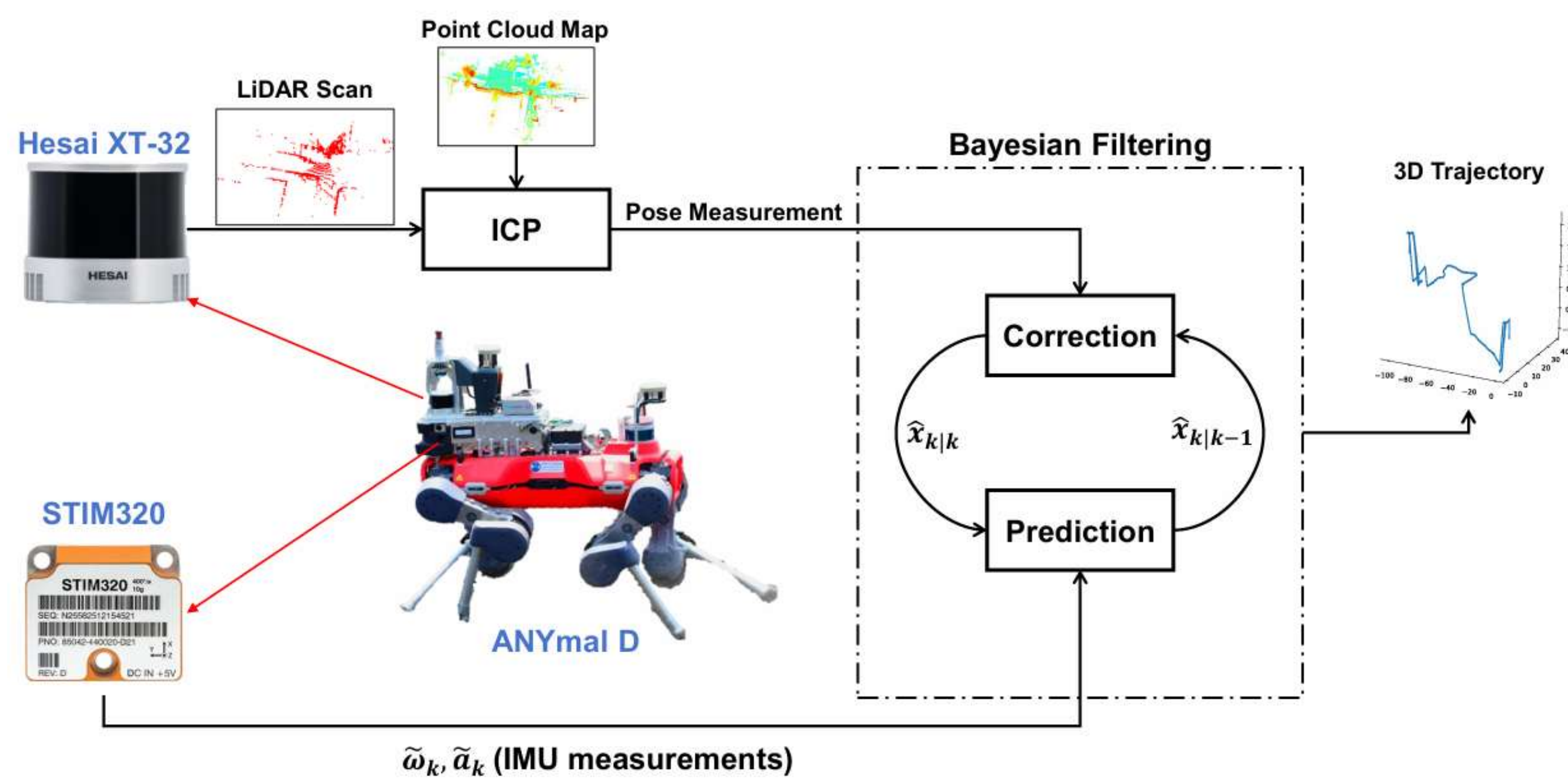
Motivation

- Accurate state estimation is a cornerstone challenge for legged platforms operating in complex environments due to terrain irregularities, foot slipping, and sensor noise.
- Need for a systematic evaluation of state estimation algorithms to determine the most robust strategy for specific constraints and environmental conditions.

Objectives

- Conduct a comprehensive benchmark of mainstream Bayesian filtering frameworks using the ANYmal GrandTour Dataset.
- Deeply investigate the impact of different approximation and sampling strategies on 3D localization performance in real-world scenarios.

Methodology



1 IMU-LiDAR State Estimation via Bayesian Filtering

- Frameworks:** Build the map, and then fuse IMU prediction with LiDAR ICP scan-to-map correction and compare four frameworks—EKF (first-order linearization), UKF (sigma-point nonlinear handling), PF (non-Gaussian, multi-modal), and InEKF (Lie-group-consistent error dynamics).

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Algorithm Implementation

Algorithm 1 PF-Based Radar-IMU Trajectory Fusion
Require: Map $\mathcal{M}$, IMU $\mathcal{S}_{imu}$, LiDAR $\mathcal{S}_{lidar}$, Estimator $\mathcal{T}_E$.
Ensure: Trajectory $\mathcal{T}$.
1: Init Load map $\mathcal{M}$; Align 1st scan via ICP; Init particles $\{x_i, w_i\}_{i=1}^N$.
2: while New LiDAR frame P_t arrives do
3: 1. Prediction (IMU Propagation):
4: for each particle i do
5: Integrate IMU $\mathcal{S}_{imu}$ to update (R_i, v_i, p_i) ;
6: Add process noise ϵ ;
7: end for
8: 2. Multi-Candidate ICP:
9: Select K indices $\mathcal{S}$ (Best, Mean, Random);
10: for each $j \in \mathcal{S}$ do
11: $T_{ij} \leftarrow T_{ij}^{(j)}$; $T_{ij} \leftarrow T_{ij}^{(j)}$;
12: $T_{ij} \leftarrow \text{ICP}(P_t, \mathcal{M}, T_{ij})$;
13: $s_j \leftarrow \text{Fitness} - \lambda \cdot \text{RMSE}$;
14: end for
15: $(j^*, T^*) \leftarrow \text{argmax}_j (s_j)$;
16: 3. Update & Gating:
17: for each particle i do
18: Update weight $w_i \propto w_i \cdot \mathcal{P}(s_i; T^*, \Sigma)$;
19: end for
20: Normalize w_i ;
21: end if
22: 4. Estimation & Resample:
23: Estimate $\hat{x}$ & Σ w/ w_i ;
24: if $N_{eff} < N_0$ then
25: Resample $\{x_i\}$; Apply Roughening;
26: end if
27: Store $\hat{x}$ in $\mathcal{T}$;
28: end while
29: return $\mathcal{T}$.

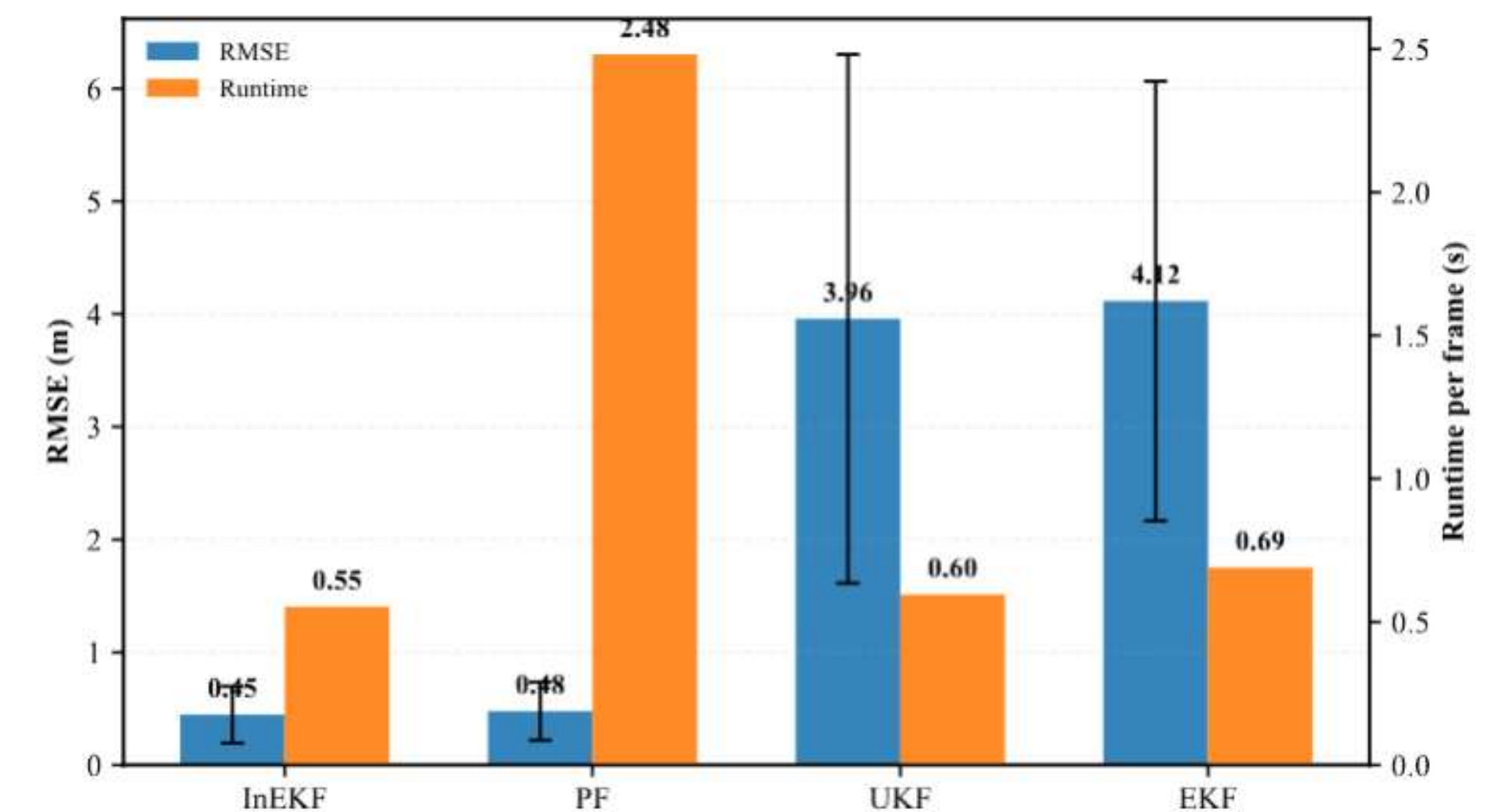
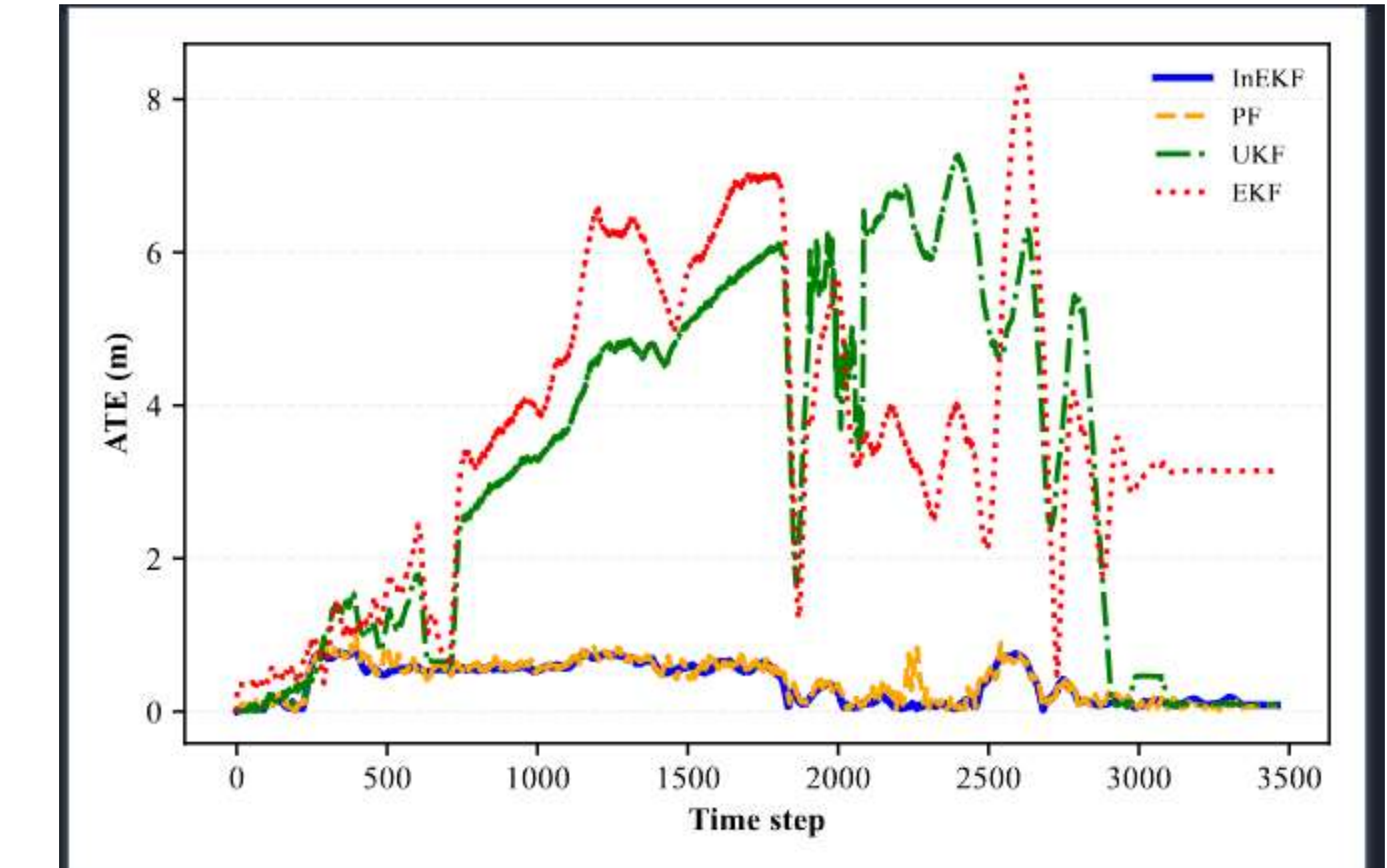
Algorithm 1 UKF-Based Radar-IMU Trajectory Fusion
Require: previous state mean μ_{k-1} , previous covariance Σ_{k-1} , IMU angular velocity ω_k , IMU linear acceleration a_k , radar position measurement z_k .
1: Estimate gyro bias, initial gravity direction, and residual acceleration bias from the initial IMU segment.
2: Integrate ω_k to update the sensor orientation and remove gravity.
3: $\hat{x}_{k-1} \leftarrow$ generate sigma points from μ_{k-1} and Σ_{k-1} .
4: $w \leftarrow$ compute the corresponding sigma-point weights.
5: for each sigma point $x_{k-1}^{(i)}$ do
6: $s_k^{(i)} \leftarrow f(x_{k-1}^{(i)}, \omega_k, \Delta t)$;
7: end for
8: $\hat{\mu}_k \leftarrow \sum_{i=1}^N w_i s_k^{(i)}$;
9: $\hat{\Sigma}_k \leftarrow \sum_{i=1}^N w_i (s_k^{(i)} - \hat{\mu}_k)(s_k^{(i)} - \hat{\mu}_k)^T + Q_k$.
10: $\hat{x}_k \leftarrow \hat{\mu}_k$;
11: Transform the radar ICP trajectory into the IMU frame using the static extrinsic calibration.
12: $\hat{\Sigma}_k \leftarrow \hat{\Sigma}_k + \sum_{i=1}^N w_i (z_k^{(i)} - \hat{z}_k)(z_k^{(i)} - \hat{z}_k)^T + R_k$.
13: $\hat{x}_k \leftarrow \hat{x}_k$;
14: $w \leftarrow$ compute the corresponding sigma-point weights.
15: for each predicted sigma point $x_k^{(i)}$ do
16: $s_k^{(i)} \leftarrow h(x_k^{(i)})$;
17: end for
18: $\hat{z}_k \leftarrow \sum_{i=1}^N w_i s_k^{(i)}$;
19: $\hat{\Sigma}_k \leftarrow \hat{\Sigma}_k + \sum_{i=1}^N w_i (s_k^{(i)} - \hat{z}_k)(s_k^{(i)} - \hat{z}_k)^T + R_k$.
20: $\hat{x}_k \leftarrow \hat{x}_k$;
21: $\hat{\Sigma}_k \leftarrow \hat{\Sigma}_k$;
22: $\hat{\mu}_k \leftarrow \hat{\mu}_k + K_k (z_k - \hat{z}_k)$;
23: $\hat{\Sigma}_k \leftarrow \hat{\Sigma}_k - K_k S_k K_k^T$;
24: $\hat{x}_k \leftarrow \hat{x}_k$;
25: return $\hat{\mu}_k, \hat{\Sigma}_k$.

Algorithm 1 EKF Algorithm
Require: previous state mean μ_{k-1} , previous covariance Σ_{k-1} , control input u_k , measurement z_k , process noise covariance Q_k , measurement noise covariance R_k , nonlinear state transition $f(\cdot)$, nonlinear measurement $h(\cdot)$.
1: Predict the state mean.
2: $\hat{\mu}_k \leftarrow f(\mu_{k-1}, u_k)$.
3: Compute the state transition Jacobian.
4: $F_k \leftarrow \frac{\partial f(\mu_{k-1})}{\partial x}$;
5: Predict the covariance.
6: $\hat{\Sigma}_k \leftarrow F_k \Sigma_{k-1} F_k^T + Q_k$.
7: Update (Measurement Update).
8: $\hat{z}_k \leftarrow h(\hat{\mu}_k)$.
9: Compute the measurement Jacobian.
10: $H_k \leftarrow \frac{\partial h(\hat{\mu}_k)}{\partial x}$.
11: Compute the innovation (residual).
12: $y_k \leftarrow z_k - \hat{z}_k$.
13: Compute the innovation covariance.
14: $S_k \leftarrow H_k \hat{\Sigma}_k H_k^T + R_k$.
15: Compute the Kalman gain.
16: $K_k \leftarrow \hat{\Sigma}_k H_k^T S_k^{-1}$.
17: Update the state mean.
18: $\hat{\mu}_k \leftarrow \hat{\mu}_k + K_k y_k$.
19: Update the covariance.
20: $\hat{\Sigma}_k \leftarrow (\hat{\Sigma}_k - K_k H_k) S_k^{-1} (I - K_k H_k)^T + K_k R_k K_k^T$.
21: $\hat{\Sigma}_k \leftarrow \hat{\Sigma}_k - (I - K_k H_k) S_k^{-1} (I - K_k H_k)^T$.
22: return $\hat{\mu}_k, \hat{\Sigma}_k$.

Algorithm 1 RIEKF (Right-Invariant Error EKF)
Require: previous estimate $R_{k-1}, p_{k-1}, v_{k-1}, b_{k-1}, \hat{g}_{k-1}$, previous covariance P_{k-1} , IMU angular velocity ω_k , IMU linear acceleration a_k , measurement z_k (e.g., radar position), time step Δt .
1: Estimate gyro bias, initial gravity direction, and residual acceleration bias from the initial IMU segment.
2: $\hat{\omega}_k \leftarrow \omega_k - b_{k-1}$; $\hat{a}_k \leftarrow a_k - \hat{g}_{k-1}$.
3: $R_k \leftarrow R_{k-1} \exp(\hat{\omega}_k^* \Delta t)$.
4: $v_k^* \leftarrow R_k v_{k-1} - g$.
5: $p_k \leftarrow p_{k-1} + v_k^* \Delta t + \frac{1}{2} \hat{a}_k^* \Delta t^2$.
6: $\Phi_k \leftarrow$ from RIEKF error dynamics.
7: $P_k \leftarrow \Phi_k P_{k-1} \Phi_k^T + Q_k$.
8: $p_k^* \leftarrow R_k p_k + p$.
9: $\hat{z}_k \leftarrow z_k - p_k^*$.
10: $H_k \leftarrow \frac{\partial h(\hat{z}_k)}{\partial x}$.
11: $S_k \leftarrow H_k P_k H_k^T + R_k$.
12: $K_k \leftarrow P_k H_k^T S_k^{-1}$.
13: $\hat{p}_k \leftarrow p_k + K_k \hat{z}_k$.
14: $\hat{b}_{k+1} \leftarrow \hat{b}_k + \hat{p}_k$.
15: $\hat{g}_{k+1} \leftarrow \hat{g}_k + \hat{p}_k$.
16: $\hat{v}_{k+1} \leftarrow \hat{v}_k + \hat{p}_k$.
17: $\hat{a}_{k+1} \leftarrow \hat{a}_k + \hat{p}_k$.
18: $\hat{z}_{k+1} \leftarrow \hat{z}_k + \hat{p}_k$.
19: $\hat{p}_{k+1} \leftarrow (I - K_k H_k) P_k (I - K_k H_k)^T + K_k R_k K_k^T$.
20: $\hat{P}_{k+1} \leftarrow (I - K_k H_k) P_k (I - K_k H_k)^T$.
21: return $R_k, p_k, v_k, b_k, \hat{g}_k, \hat{v}_k, \hat{a}_k, \hat{z}_k, P_k$.

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Error and Runtime comparison



Discussion

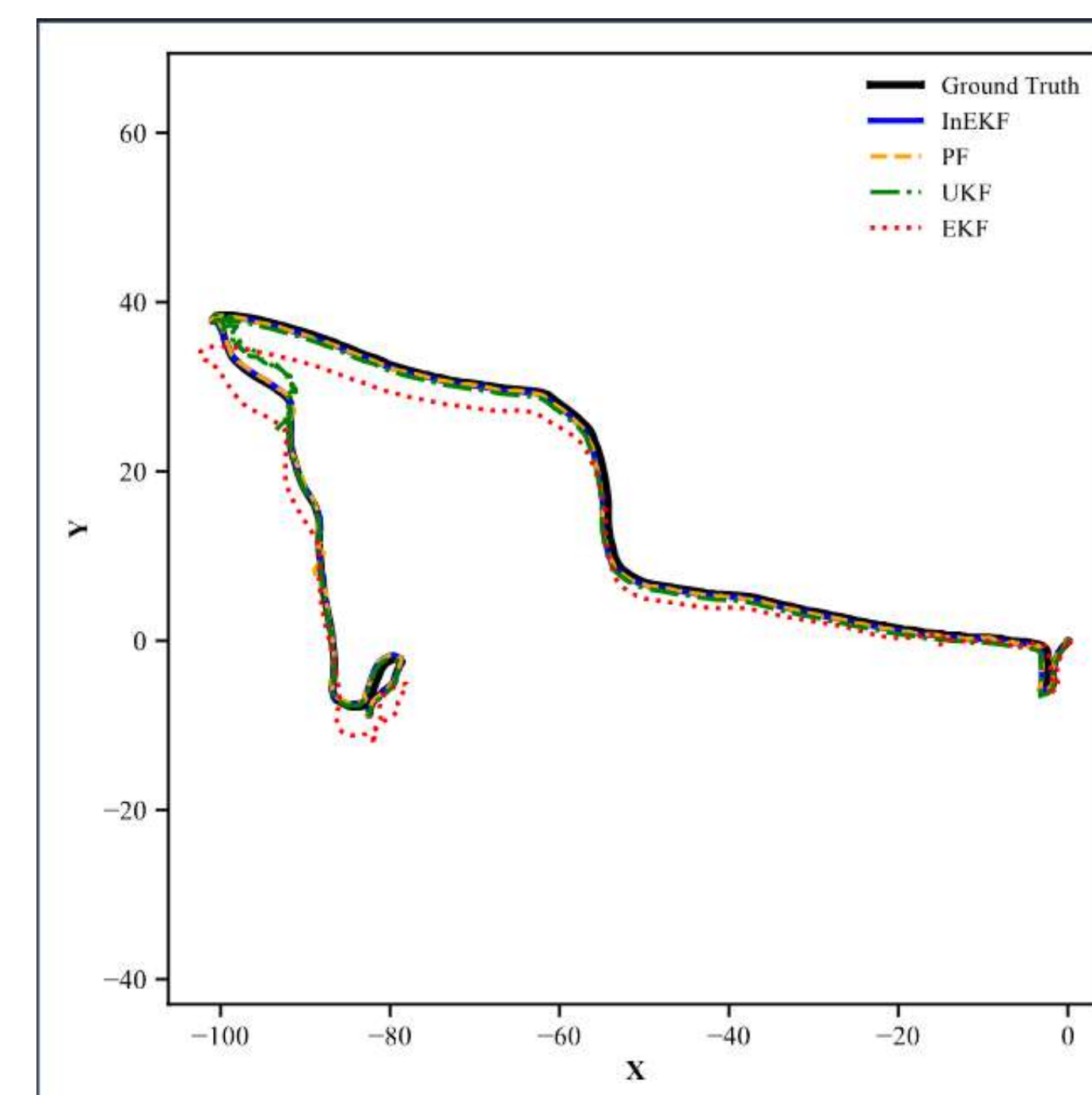
- Our approach achieves high performance with real-time capability, especially regarding the In-EKF filter.
- All methods perform similarly in simple environments, but InEKF shows better robustness in challenging scenarios, while PF is more computationally expensive.
- The main limitation is the reliance on ICP measurements, which can lead to drift when the measurements become unreliable.

Future Works

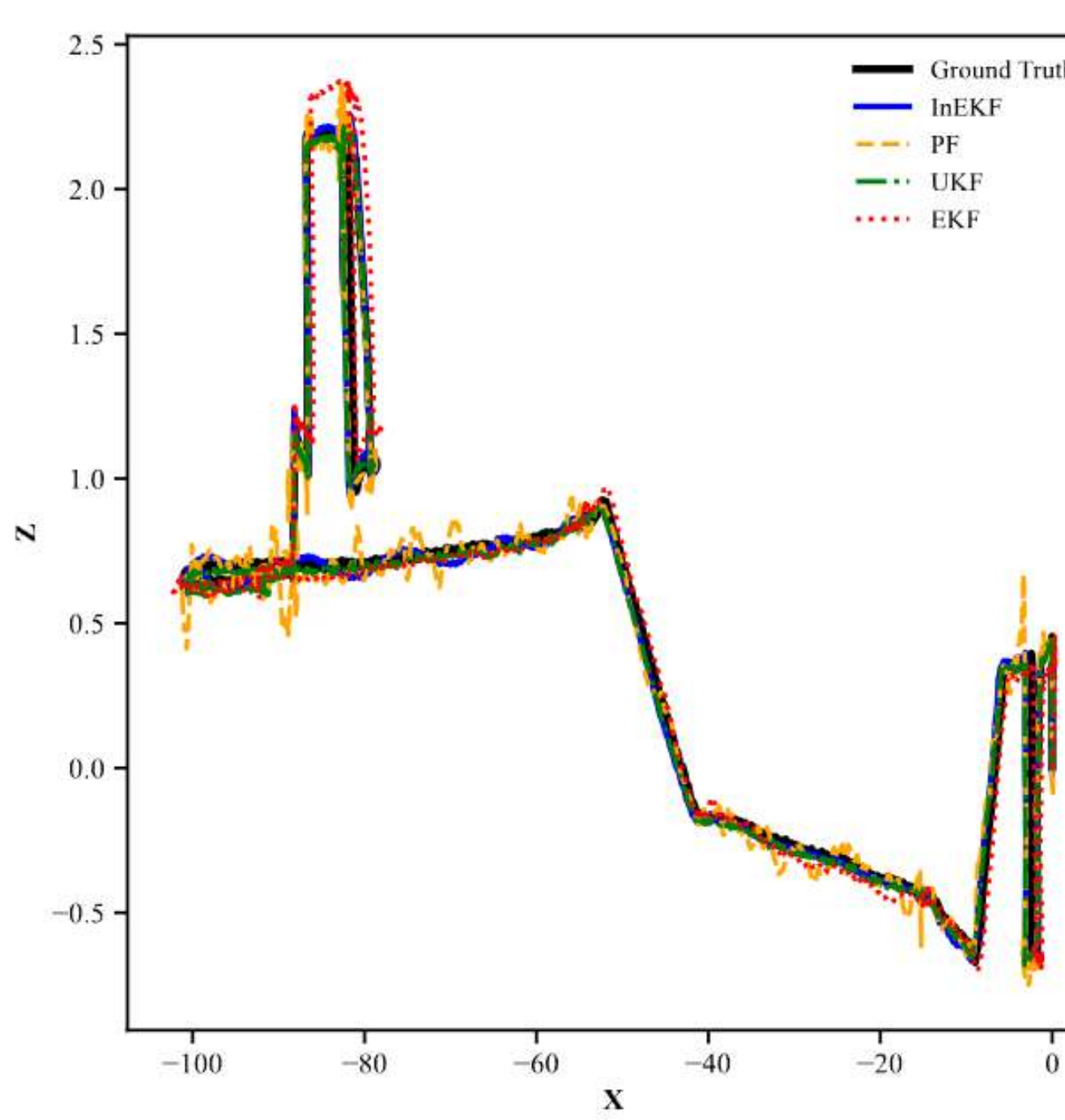
- Future work will focus on improving robustness under perceptual degeneracy.
- Another promising direction is to incorporate contact information within an invariant filtering framework.

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Trajectory Comparison



Top view



Side view

References: [1] J. Frey, T. Tuna, F. Fu, K. Patterson, T. Xu, M. Fallon, C. Cadena, and M. Hutter, "GrandTour: A Legged Robotics Dataset in the Wild for Multi-Modal Perception and State Estimation," arXiv:2602.18164, 2026.

[2] J. Frey, T. Tuna, L. F. T. Fu, C. Weibel, K. Patterson, B. Krummenacher, M. Müller, J. Nubert, M. Fallon, C. Cadena, and M. Hutter, "Boxi: Design Decisions in the Context of Algorithmic Performance for Robotics," in *Proc. Robotics: Science and Systems*, Los Angeles, USA, Jul. 2025.

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